Functional Requirements Document – Employee Workload

TaskFlow – Intelligent Task Allocation & Tracking System

> **Document:** Functional Requirements Document (sub-module 02.2 Employee Workload, code `WORK`) · **Version:** 1.0 · **Status:** Draft **Project type:** Personal Project (Skill Development) · **Date:** 31-Oct-2026 **Author:** Jojo George (jojogeorge3344@gmail.com) **Based on:** BRD-Employee-Workload.md v1.0 © 2026 Jojo George. All rights reserved.

Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version, derived from BRD-Employee-Workload v1.0 and the verified Employee Workload workflow doc

Related Documents

Document	Purpose	Location
BRD – Employee Workload	Business needs this FRD details (`REQ-WORK-nn`, `BR-WORK-nn`)	`../../../BRD/02-TaskFlow-Monitoring/02.2-Employee-Workload/BRD-Employee-Workload.md`
Overall BRD / FRD	Roles, global rules `BR-nn`, global NFRs	`../../../BRD/00-Overall/BRD-TaskFlow-Overall.md`, `../../00-Overall/FRD-TaskFlow-Overall.md`
Sibling FRDs – Task Management, Board	Target of the **Tasks** → link; task data and definitions	`../02.1-Task-Management/FRD-Task-Management.md`, `../02.3-Board/FRD-Board.md`
Development Plan	Screen-wise build plan (original Screen 16)	`TF - Documents/Screens Wise Feature Development Plan/16-employee-workload-screen.md`
HLD / LLD / TDD	Technical design	`../../../HLD/02-TaskFlow-Monitoring/02.2-Employee-Workload/HLD-Employee-Workload.md`, `../../../LLD/02-TaskFlow-Monitoring/02.2-Employee-Workload/LLD-Employee-Workload.md`, `TDD-TaskFlow.md` (v2.0)

1. Introduction

1.1 Purpose

This FRD specifies the detailed, testable behaviour of the Employee Workload screen: which people are listed, how each count is worked out, how overloaded people are flagged, how the filters behave, the link to a person's tasks,

and who may use each part.

1.2 Scope

In scope: the Employee Workload screen (`/workload`) – per-employee task counts, project allocations, the average and "High load" highlighting, the Project and Employee filters and the **Tasks** → link.

Out of scope (see BRD §3.2): a Department column or filter; a separate workload chart (workload charts are on the Dashboard); changing or reassigning tasks on this screen (Task Management, `../02.1-Task-Management/FRD-Task-Management.md`); narrowing the list to a Manager's own team.

1.3 Audience

Developer, tester and reviewer of TaskFlow (Jojo George), and the product owner approving the behaviour.

1.4 Conventions

- Requirement IDs: `FR-WORK-nn` . Use cases: `UC-WORK-nn` . BRD requirements: `REQ-WORK-nn` ; sub-module rules `BR-WORK-nn` ; global rules `BR-nn` (overall BRD).
- Priority: **Must / Should / Could** (MoSCoW).
- "The system shall …" = mandatory behaviour; "should" = recommended.
- Text in quotes is the exact UI label or message; `{name}` marks a value filled in at run time.

1.5 Definitions & Abbreviations

Term	Meaning
Listed employee	An active user whose role does not hold `tasks.view.all` (the Employee role by default)
Active task	A task that is not Completed, Cancelled or deleted
High priority task	An active task with High or Critical priority
Overdue task	An active task whose deadline has passed
Allocation	A person's percentage share of a project, with a position
Allocated %	The sum of a person's project allocation percentages
Average	Total active tasks of the rows shown divided by the number of rows shown
High load	Active count at least twice the average of the rows shown
Task scope	The tasks a user may see: Own, Team or Organization

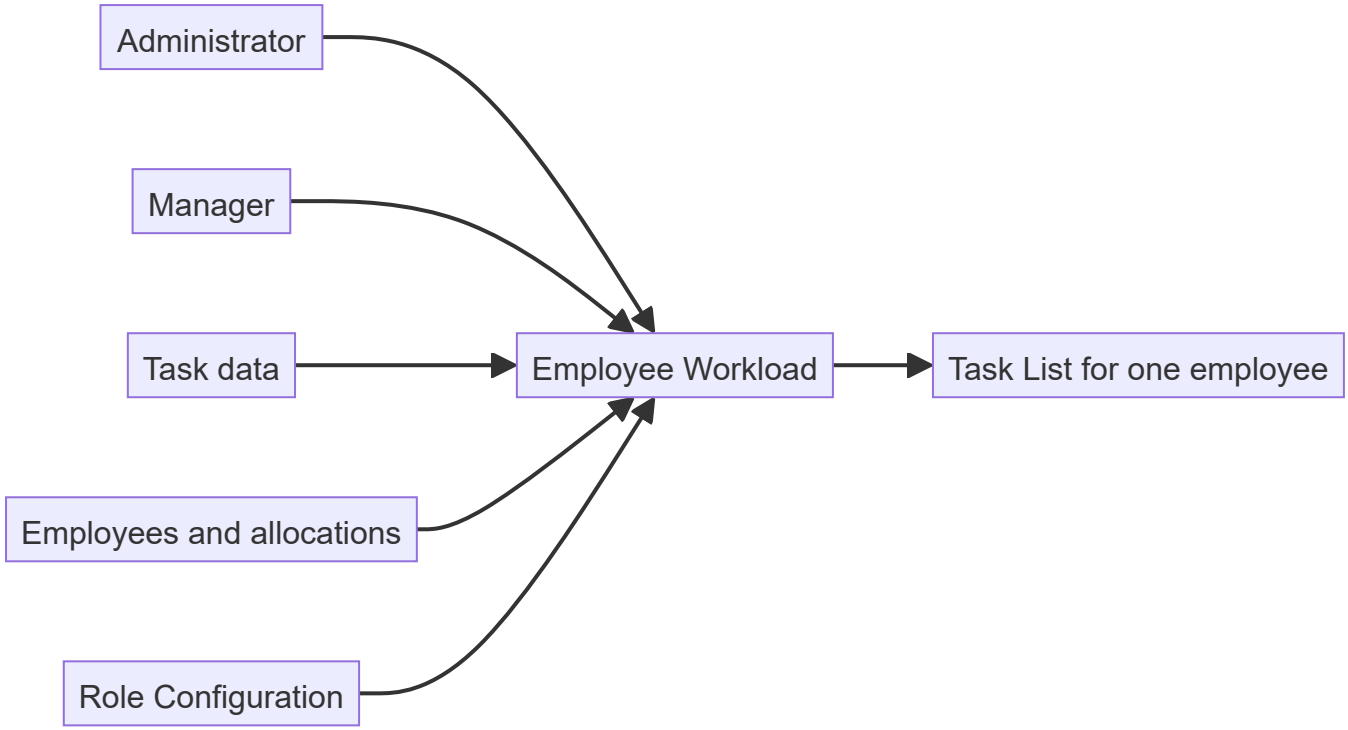
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2. System Overview

2.1 Module Map

Code	Module	Child modules / screens	Primary roles
WORK	Employee Workload (TaskFlow Monitoring)	Employee Workload	Administrator, Manager
TASK	Task Management (related)	Task List – target of **Tasks** →; source of the task data	All users
EMP / PROJ	Employee and Project Management (related)	Employees and project allocations	Administrator, Manager
ROLE	Role Configuration (related)	Grants the workload keys; which roles hold `tasks.view.all` decides who is listed	Administrator

2.2 Context Diagram



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3. User Roles & Permissions

3.1 Roles

Role	Description	Data scope
Administrator	Reviews load across the organization, filters, opens an employee's tasks	Organization
Manager	Reviews load, filters, opens an employee's tasks to rebalance work	Organization on this screen (not narrowed to the team); the Task List opened from it keeps the Manager's normal task scope
Employee	Is listed on the screen; cannot open it by default	No access
Custom role	As granted by `workload.view`, `workload.filters.use`, `workload.tasks.open`	Organization when granted

3.2 Permission Matrix (functional view)

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Employee Workload › View workload	`workload.view`	Granted	Granted	Denied	Configurable
Employee Workload › Use workload filters (requires `workload.view`)	`workload.filters.use`	Granted	Granted	Denied	Configurable
Employee Workload › Open an employee's tasks (requires `workload.view`)	`workload.tasks.open`	Granted	Granted	Denied	Configurable

Module › Function	Permission key	Administrator	Manager	Employee	Custom role
Task Management › View all tasks (decides who is listed; not used to open the screen)	`tasks.view.all`	Granted	Granted	Denied	Configurable
Project Management › View projects (fills the Project filter)	`projects.view`	Granted	Granted	Denied	Configurable

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4. Functional Requirements by Module

4.1 Module: Employee Workload (WORK)

Purpose: Show how open work is spread across employees so Administrators and Managers can spot overload and overdue work and rebalance it. · **BRD reference:** BRD-Employee-Workload §5 SCREEN 1, BR-WORK-01…06; global BR-04, BR-06, BR-12, BR-22.

4.1.1 Screen: Employee Workload

Field	Value
Route / location	`/workload` – menu **TaskFlow Monitoring → Employee Workload**
Access	`workload.view` (Administrator, Manager; Custom role as granted). Employee: no access by default
Entry points	Menu item (shown only with `workload.view`)

Screen elements

Element	Type	Description	Mandatory	Visible to
Header summary	Text	"N employees · average X active tasks" (average with one decimal; "1 employee" in the singular)	–	`workload.view`
Project	Drop-down	Narrows to people allocated to the project; reloads the list	No	`workload.filters.use`
Employee	Drop-down	Shows one person; applies on screen without reloading	No	`workload.filters.use`
Reset	Button	Clears both filters and reloads	–	`workload.filters.use`
Employee	Column	Name with a "High load" badge when applicable	–	`workload.view`
Projects	Column	Each project name and allocation %	–	`workload.view`
Allocated %	Column	Sum of the allocations	–	`workload.view`
Active, High priority, Overdue, Completed	Columns	Task counts; Overdue highlighted when greater than 0	–	`workload.view`
Workload bar	Bar	Active count scaled to the busiest row shown	–	`workload.view`

Element	Type	Description	Mandatory	Visible to
Tasks →	Link	Opens the Task List filtered to the employee	–	`workload.tasks.open`
Legend	Text	"Highlighted rows carry at least twice the average active load (X active tasks)."	–	`workload.view`

Functional requirements

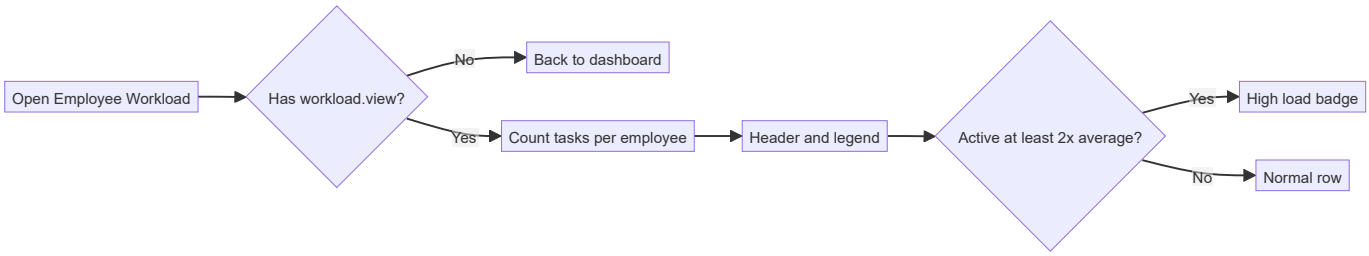
ID	Requirement	Priority	Permission	BRD trace
FR-WORK-01	The system shall list every active user whose role does not hold `tasks.view.all` (the Employee role by default), ordered by name.	Must	`workload.view`	REQ-WORK-01, BR-WORK-02
FR-WORK-02	The system shall never list inactive users.	Must	`workload.view`	BR-WORK-02
FR-WORK-03	The system shall show each employee's Active count: assigned tasks that are not Completed, Cancelled or deleted.	Must	`workload.view`	REQ-WORK-02, BR-WORK-01
FR-WORK-04	The system shall show each employee's High priority count: active tasks with priority High or Critical.	Must	`workload.view`	REQ-WORK-03, BR-WORK-01
FR-WORK-05	The system shall show each employee's Overdue count (active tasks whose deadline has passed) and highlight it when greater than 0.	Must	`workload.view`	REQ-WORK-04, BR-WORK-01, BR-12
FR-WORK-06	The system shall show each employee's Completed count (tasks with status Completed that are not deleted).	Must	`workload.view`	REQ-WORK-05, BR-WORK-01
FR-WORK-07	The system shall list employees who have no tasks with zero in every count.	Should	`workload.view`	REQ-WORK-06
FR-WORK-08	The system shall show each employee's project allocations (project name and percentage) and an Allocated % equal to their sum.	Should	`workload.view`	REQ-WORK-07, BR-WORK-04
FR-WORK-09	The system shall show in the header the number of rows shown and their average active tasks, "N employees · average X active tasks".	Should	`workload.view`	REQ-WORK-08
FR-WORK-10	The system shall mark with "High load" every row whose active count is at least twice the average of the rows currently shown, and mark no row when that average is 0.	Must	`workload.view`	REQ-WORK-09, BR-WORK-03
FR-WORK-11	The system shall explain the highlight in the legend "Highlighted rows carry at least twice the average active load (X active tasks)."	Must	`workload.view`	REQ-WORK-09
FR-WORK-12	The system shall show a workload bar per row, scaled so the busiest row shown is full length.	Could	`workload.view`	REQ-WORK-10

ID	Requirement	Priority	Permission	BRD trace
FR-WORK-13	The system shall, when a Project is picked, reload the list with only the people allocated to that project and clear the Employee filter.	Should	`workload.filters.use`	REQ-WORK-11
FR-WORK-14	The system shall, when an Employee is picked, show only that person without reloading, recalculate the header, average, highlight and bars for the rows shown, and offer **Reset** to clear both filters.	Should	`workload.filters.use`	REQ-WORK-12, NFR-WORK-01
FR-WORK-15	The system shall open the Task List filtered to the row's employee when **Tasks** → is clicked, applying the user's own task scope.	Must	`workload.tasks.open`	REQ-WORK-14, BR-WORK-05
FR-WORK-16	The system shall show the Project and Employee filters and **Reset** only to users with `workload.filters.use`.	Should	`workload.filters.use`	REQ-WORK-13, BR-WORK-06
FR-WORK-17	The system shall show the **Tasks** → link only to users with `workload.tasks.open`.	Must	`workload.tasks.open`	REQ-WORK-14, BR-WORK-06
FR-WORK-18	The system shall show the menu item and open the screen only for users with `workload.view`, send everyone else back to their dashboard and refuse the workload data to a direct request without the key.	Must	`workload.view`	REQ-WORK-15, BR-WORK-06, NFR-WORK-02
FR-WORK-19	The system shall cover the whole organization for every user who may open the screen, not only a Manager's team.	Must	`workload.view`	REQ-WORK-16
FR-WORK-20	The system shall not create, change or delete any data from this screen; viewing and filtering are read-only.	Should	`workload.view`	REQ-WORK-17, NFR-WORK-04
FR-WORK-21	The system shall show "No employees to show – There are no active employees in this project." when no rows match, and "Could not load the workload. Check your connection and try again." when loading fails.	Should	`workload.view`	REQ-WORK-01, REQ-WORK-11
FR-WORK-22	The system shall use the same active, overdue and priority definitions as the Dashboard, Task List and Reports.	Should	`workload.view`	BR-WORK-01, NFR-WORK-05, BR-22

Use case: UC-WORK-01 – Review workload

Item	Detail
Actor	Manager or Administrator
Preconditions	Signed in; holds `workload.view`
Trigger	Opens **TaskFlow Monitoring → Employee Workload**
Main flow	1. System loads every listed employee with their allocations and four counts. 2. System shows the header "N employees · average X active tasks" and the legend. 3. System marks rows at or above twice the average with "High load" and highlights Overdue counts above 0. 4. User reviews the table and bars.

Item	Detail
Alternate flows	A1: No listed employees → "No employees to show – There are no active employees in this project."
Exceptions	E1: Load failure → "Could not load the workload. Check your connection and try again." E2: No `workload.view` → menu hidden; the screen sends the user to the dashboard; data refused.
Postconditions	Nothing changed



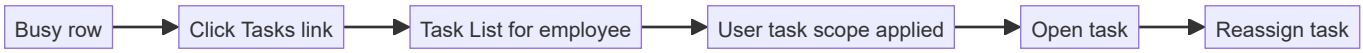
Use case: UC-WORK-02 – Filter by project or employee

Item	Detail
Actor	Manager or Administrator
Preconditions	Holds `workload.view` and `workload.filters.use`
Trigger	Picks a Project or an Employee
Main flow	1. Project picked: system reloads with only the people allocated to it and clears the Employee filter. 2. Employee picked: system shows only that person without reloading. 3. System recalculates header, average, "High load" and bars for the rows shown. 4. **Reset** clears both filters and reloads.
Alternate flows	–
Exceptions	E1: Nobody allocated to the project → "No employees to show – There are no active employees in this project."
Postconditions	Nothing changed

Use case: UC-WORK-03 – Open an employee's tasks

Item	Detail
Actor	Manager or Administrator
Preconditions	Holds `workload.view` and `workload.tasks.open`
Trigger	Clicks **Tasks** → on a row
Main flow	1. System opens the Task List filtered to that employee. 2. The Task List applies the user's own task scope (a Manager sees only team tasks plus own). 3. User opens a task and reassigns it (Task Management, UC-TASK-09).
Alternate flows	–

Item	Detail
Exceptions	E1: The employee has no tasks within the user's scope → the Task List shows its empty state.
Postconditions	Work can be rebalanced in Task Management



Field validations

Field	Rule	Error message
Project	Optional; one of the projects the user may view	–
Employee	Optional; one of the rows loaded	–

Business rules applied: BR-WORK-01, BR-WORK-02, BR-WORK-03, BR-WORK-04, BR-WORK-05, BR-WORK-06; global BR-04, BR-06, BR-12, BR-22.

Notifications / side effects

Event	Who is notified	Channel
–	Read-only screen; nothing is sent	–

Messages

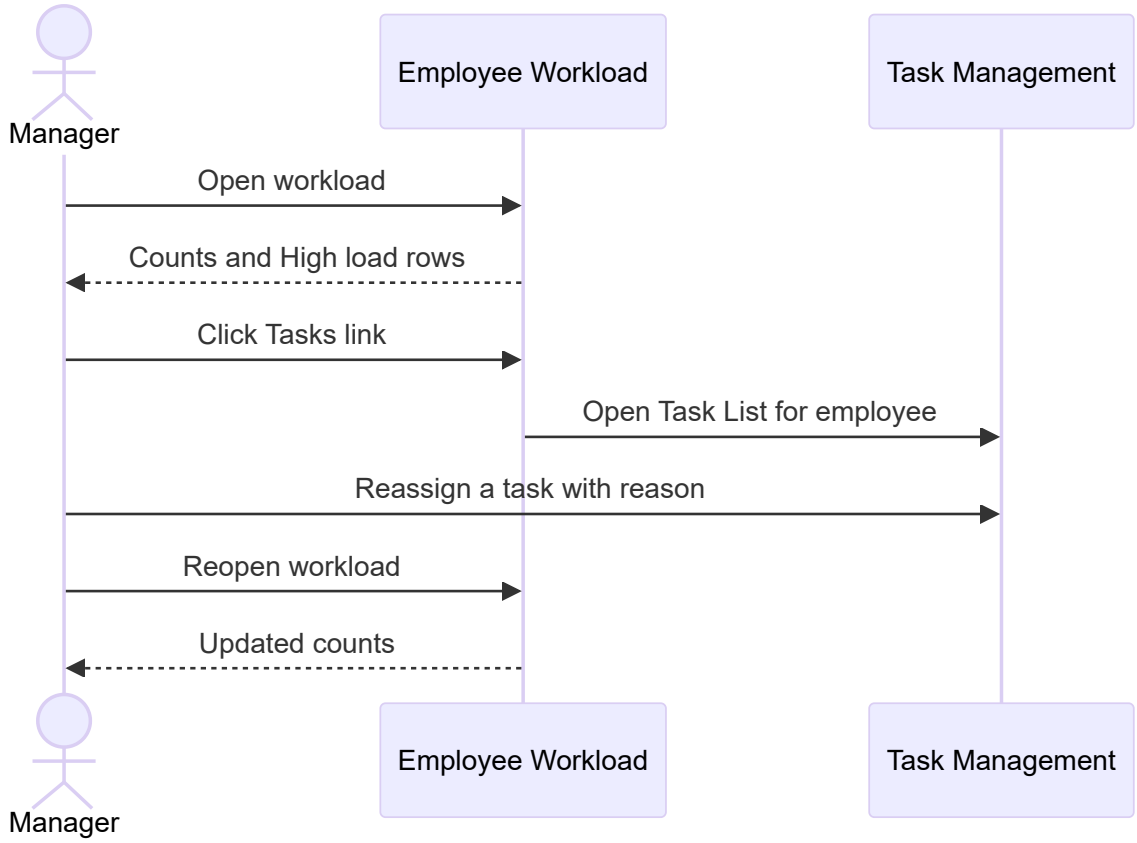
Situation	Message
Load failed	"Could not load the workload. Check your connection and try again."
No rows	"No employees to show – There are no active employees in this project."
Legend	"Highlighted rows carry at least twice the average active load (X active tasks)."
No permission	Menu item hidden; the screen sends the user to their dashboard; the data is refused

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5. Cross-Module Workflows

5.1 Spot overload and rebalance work

Step	Module	Actor / System	Action	FR reference
1	WORK	Manager	Opens Employee Workload and sees a "High load" row	FR-WORK-01…FR-WORK-11
2	WORK	Manager	Clicks **Tasks** → on that row	FR-WORK-15, FR-WORK-17
3	TASK	System	Task List filtered to the employee within the Manager's scope	FR-TASK-01, FR-TASK-05 (Task Management FRD)
4	TASK	Manager	Reassigns a task with a reason	FR-TASK-48…FR-TASK-55 (Task Management FRD)
5	WORK	Manager	Reopens Employee Workload; the counts reflect the move	FR-WORK-03



6. Reports & Exports

Not applicable for this sub-module. Workload is an on-screen view with no export; exportable reports are in `../../05-Reports/05.1-Reports/FRD-Reports.md`.

7. Notifications & Messages Catalogue

Not applicable for this sub-module. The screen is read-only and sends no notifications or mail. On-screen messages are listed in §4.1.1.

8. Data Requirements (functional view)

This sub-module stores no data of its own; it reads the items below.

Data item	Description	Created by	Rules
Listed employee	Name, active flag, role	Employee Management	Only active users whose role lacks `tasks.view.all`
Project allocation	Project, position, allocation %	Employee / Project Management	Projects listed by project name; Allocated % = sum
Task counts	Active, High priority, Overdue, Completed per assignee	System (on request)	Deleted tasks never counted; Overdue uses the current time
Average / High load	Average active of the rows shown; badge at ≥ 2 × average	System (on screen)	Recalculated whenever the rows shown change; no badge when the average is 0

9. Non-Functional Requirements (functional impact)

ID	Requirement	Affects
NFR-WORK-01	One load per screen open or Project change; the Employee filter works without reloading	Employee Workload (FR-WORK-13, FR-WORK-14)
NFR-WORK-02	Workload data refused without `workload.view`	Employee Workload (FR-WORK-18)
NFR-WORK-03	Overload and overdue work stand out: "High load" badge, highlighted Overdue, legend	Employee Workload (FR-WORK-05, FR-WORK-10, FR-WORK-11)
NFR-WORK-04	Read-only screen	Employee Workload (FR-WORK-20)
NFR-WORK-05	Same count definitions as Dashboard and Reports	Employee Workload (FR-WORK-22)
NFR-WORK-06 / NFR-WORK-07	Compatibility and accessibility as the global NFRs (NFR-16, NFR-18)	Employee Workload

10. Assumptions & Open Issues

Type	Description	Owner
Assumption	Project allocations are kept up to date in Employee and Project Management	Jojo George
Assumption	Users who open the screen also hold `projects.view`, so the Project filter can be filled (true for Administrator and Manager by default)	Jojo George
Open issue	`workload.filters.use` and `workload.tasks.open` are enforced only on the screen; the Project filter works by direct request without `workload.filters.use` (R-WORK-01)	Jojo George
Open issue	A Manager sees counts of employees outside the team (accepted for this release, FR-WORK-19; R-WORK-02)	Jojo George
Open issue	A role that holds `tasks.view.all` only through an older permission name could still be listed (R-WORK-03)	Jojo George
Open issue	No Department column or filter, unlike the original BRD Screen 16 (R-WORK-04)	Jojo George
Open issue	"High load" is relative to the rows shown, so picking a single Employee never shows the badge	Jojo George

11. Requirements Traceability Matrix

Test cases name the automated test files (`TaskFlow.UnitTests` / `TaskFlow.IntegrationTests` and the frontend `*.spec.ts` files under `frontend/src/app/`) that test the behaviour; the mapping was checked against the test sources. "Manual UI" = checked by hand on the screen.

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-WORK-01	FR-WORK-01, FR-WORK-21	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`, `CustomRoleWorkloadTests.cs`, `workload.component.spec.ts` (error state; the empty state is not tested)

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-WORK-02	FR-WORK-03	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-03	FR-WORK-04	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-04	FR-WORK-05	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-05	FR-WORK-06	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-06	FR-WORK-07	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-07	FR-WORK-08	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`, `workload.component.spec.ts`
REQ-WORK-08	FR-WORK-09	Employee Workload	UC-WORK-01	`workload.component.spec.ts` (row count; the average is not asserted)
REQ-WORK-09	FR-WORK-10, FR-WORK-11	Employee Workload	UC-WORK-01	`workload.component.spec.ts` (highlight; the legend text is not asserted)
REQ-WORK-10	FR-WORK-12	Employee Workload	UC-WORK-01	`workload.component.spec.ts` (bar shown; the scaling is not asserted)
REQ-WORK-11	FR-WORK-13, FR-WORK-21	Employee Workload	UC-WORK-02	`WorkloadEndpointsTests.cs`, `WorkloadServiceTests.cs`, `workload.component.spec.ts`
REQ-WORK-12	FR-WORK-14	Employee Workload	UC-WORK-02	`workload.component.spec.ts`
REQ-WORK-13	FR-WORK-16	Employee Workload	UC-WORK-02	`workload.component.spec.ts`

BRD item	FRD requirement(s)	Screen	Use case	Test case
REQ-WORK-14	FR-WORK-15, FR-WORK-17	Employee Workload	UC-WORK-03	`workload.component.spec.ts`, `task-list.component.spec.ts`
REQ-WORK-15	FR-WORK-18	Employee Workload	UC-WORK-01	`WorkloadEndpointsTests.cs`, `app.routes.spec.ts`
REQ-WORK-16	FR-WORK-19	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
REQ-WORK-17	FR-WORK-20	Employee Workload	UC-WORK-01, UC-WORK-02	Manual UI
BR-WORK-01	FR-WORK-03, FR-WORK-04, FR-WORK-05, FR-WORK-06, FR-WORK-22	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`
BR-WORK-02	FR-WORK-01, FR-WORK-02	Employee Workload	UC-WORK-01	`CustomRoleWorkloadTests.cs`
BR-WORK-03	FR-WORK-10	Employee Workload	UC-WORK-01, UC-WORK-02	`workload.component.spec.ts`
BR-WORK-04	FR-WORK-08	Employee Workload	UC-WORK-01	`WorkloadServiceTests.cs`, `workload.component.spec.ts`
BR-WORK-05	FR-WORK-15	Employee Workload	UC-WORK-03	`TasksEndpointsTests.cs`, `task-list.component.spec.ts`, `workload.component.spec.ts`
BR-WORK-06	FR-WORK-16, FR-WORK-17, FR-WORK-18	Employee Workload	UC-WORK-01, UC-WORK-02, UC-WORK-03	`WorkloadEndpointsTests.cs`, `workload.component.spec.ts`, `app.routes.spec.ts`

12. Approvals

Role	Name	Decision	Date
Product owner	Jojo George	Pending	–
Reviewer	Jojo George	Pending	–

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